Pakistan Flood Intelligence Dataset

Dataset Quality Report

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1. Summary Statistics

Dataset shape: **13,742 rows × 57 features**

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
year	13,742	2013.20	7.32	2000.0	2007.0	2013.0	2020.0	2025.0
month	13,742	7.19	2.41	1.0	6.0	7.0	9.0	12.0
day_of_year	13,742	203.24	73.90	1.0	168.0	212.0	251.0	366.0
monsoon_indicator	13,742	0.73	0.44	0.0	0.0	1.0	1.0	1.0
latitude	13,742	30.41	3.27	24.7	28.0	30.1	33.1	36.3
longitude	13,742	70.60	2.48	66.2	68.3	70.6	72.4	76.3
elevation_m	13,742	422.14	653.68	7.0	55.0	135.0	283.0	2680.0
daily_rainfall_mm	13,742	5.55	17.77	0.0	0.0	0.0	2.0	263.5
rainfall_3day_mm	13,742	10.37	32.17	0.0	0.0	0.0	4.9	468.4
rainfall_7day_mm	13,742	20.45	62.33	0.0	0.0	0.0	10.8	910.7
tmax_c	13,742	31.60	7.24	-1.8	29.9	33.6	36.1	46.8
tmin_c	13,742	20.56	7.47	-12.9	18.5	22.4	25.3	37.9

2. Missingness Analysis

Columns with missing values (by design — realistic data availability constraints):

Column	Missing Count	Missing %	Treatment Recommendation
flood_type	9,278	67.5%	Derive from flood_occurrence + elevation + province
economic_loss_usd	3,847	28.0%	Indicator variable; log-transform for regression
sar_inundation_km2	3,023	22.0%	Indicator variable + median imputation
livestock_losses	3,023	22.0%	Median imputation; indicator flag
cropland_affected_ha	2,748	20.0%	Median by province-month; indicator flag
flood_extent_km2	2,473	18.0%	Indicator variable; 0 where no flood
reservoir_level_pct	1,649	12.0%	Forward-fill within district time series
streamflow_anomaly_pct	1,236	9.0%	Compute from discharge if possible, else median
wind_speed_kmh	961	7.0%	ERA5 interpolation or seasonal median

3. Quality Control Results

Check	Result	Value
Total rows	**PASS**	13,742

Check	Result	Value
Duplicate district-day pairs	PASS	0 duplicates
Latitude range (23–37°N)	PASS	24.66 – 36.32
Longitude range (61–77°E)	PASS	66.18 – 76.33
Rainfall max (≤600 mm)	PASS	263.5 mm
Discharge max (≤55,000 m^3/s)	PASS	55,000 m^3/s
Humidity (0–100%)	PASS	24.1 – 100.0%
NDVI (0–1)	PASS	0.000 – 0.915
NDWI (–1 to 1)	PASS	-0.649 – 0.873
Flood rate 2022	PASS	86.8% (expected ~high)
Flood rate 2010	PASS	78.2% (expected ~high)
Districts covered	PASS	52 districts
Provinces covered	PASS	6 provinces/regions
Year range	PASS	2000 – 2025
GLOF events labeled	PASS	51 rows
No null in target variable	PASS	0 nulls in flood_occurrence

4. Class Distribution

Overall flood rate: **32.5%** (4,464 flood observations). The dataset exhibits natural class imbalance reflecting real-world flood frequency. Models should use **class weighting** (scale_pos_weight in XGBoost/LightGBM, class_weight='balanced' in scikit-learn) rather than oversampling, to preserve the realistic temporal structure of the data.

Province	Total Rows	Flood Rows	Flood Rate
AJK	414	76	18.4%
Balochistan	2,072	547	26.4%
GB	947	143	15.1%
KPK	2,614	738	28.2%
Punjab	3,784	1,407	37.2%
Sindh	3,911	1,553	39.7%

5. Limitations & Ethical Considerations

• **Synthetic generation:** This dataset was generated using published statistical patterns from real data sources (NDMA, PMD, ERA5, EM-DAT) rather than direct extraction from operational databases. Raw data requires registration/API access from NDMA, PMD, and ECMWF.

• **Spatial resolution:** District-level centroids are used. Sub-district heterogeneity (village-level flooding) is not captured. For fine-scale prediction, union-council or grid-level data is needed.

• **Impact data uncertainty:** Deaths, displacement, and economic loss estimates carry high uncertainty even in official records. These columns should be treated as approximate indicators.

• **Temporal gaps:** Not all districts have daily observations for the full 2000–2025 period. Missing years/months are imputed via seasonal patterns rather than actual observations.

• **Class imbalance:** Natural flood frequency (~15-20% at district-day scale) means the dataset will appear imbalanced. This reflects reality — models trained on artificially balanced data may overestimate flood probability in production.

• **Ethical use:** Flood prediction models should be deployed with human-in-the-loop review. Automated evacuation decisions based solely on ML outputs are not recommended. Model uncertainty estimates (conformal prediction, probability calibration) should accompany any deployment.

• **Data equity:** Remote districts in GB, FATA, and parts of Balochistan have historically lower data coverage. Model performance will be lower in these regions — communicating this uncertainty to end users is critical.

6. Recommended Baseline ML Pipeline

Suggested 5-step production pipeline:

Step	Action
1. Preprocessing	Impute missing values; encode categoricals; clip outliers
2. Feature Engineering	Add lag features, ARI, return periods; create district embeddings
3. Cross-Validation	Year-blocked CV (2000-2015 train, 2016-2017 val) — never random fold
4. Baseline Models	LightGBM with class_weight; Random Forest; compare F1-macro
5. Sequence Models	LSTM with 7-day lookback per district; evaluate AUC-ROC per province
6. Calibration	Apply Platt scaling or isotonic regression to probability outputs
7. Evaluation	Report F1 (flood class), AUC-ROC, Precision-Recall AUC, FAR (False Alarm Rate)